

The bound in Theorem 11.4.5 is particularly easy to compute for *balanced* graphs H , those for which $\varepsilon'(H) = \varepsilon(H)$. Cycles and trees are examples of balanced graphs. For cycles, we have the threshold familiar from the proof of Erdős's theorem:

Corollary 11.4.6. *If $k \geq 3$, then $t(n) = n^{-1}$ is a threshold function for the property of containing a k -cycle.* $\square$

Note that t does not depend on k . (See also Exercise 18.)

For trees, there is a similar phenomenon. Here, the threshold function does depend on the order of the tree, but not on its shape:

Corollary 11.4.7. *If T is a tree of order $k \geq 2$, then $t(n) = n^{-k/(k-1)}$ is a threshold function for the property of containing a copy of T .* $\square$

The systematic study of threshold functions has led to an overall picture of how the typical properties of a graph $G \in \mathcal{G}(n, p)$ unfold as the growth rate of $p = p(n)$ increases. This picture, dubbed the *evolution of random graphs*, is quite fascinating: as in the evolution of species, changes happen 'in jumps', marked by the times the growth rate of p crosses that of a threshold function.

For a very rough sketch, let us begin with edge probabilities p whose order of magnitude lies below n^{-2} ; for such p , a random graph $G \in \mathcal{G}(n, p)$ almost surely has no edges at all. But as p grows, it acquires more and more structure. As p approaches n^{-1} , its components become larger and larger trees (Corollary 11.4.7), until at $p = n^{-1}$ the first cycles are born (Exercise 18). Soon, some of these will have several crossing chords, making the graph non-planar. At the same time, one component outgrows the others, until it devours them around $p = (\log n)n^{-1}$, making the graph connected. Hardly later, at a mere $p = (1 + \epsilon)(\log n)n^{-1}$, our random graph already almost surely has a Hamilton cycle...

Exercises

1. What is the probability that a random graph in $\mathcal{G}(n, p)$ has exactly m edges, for $0 \leq m \leq \binom{n}{2}$ fixed?
2. What is the expected number of edges in $G \in \mathcal{G}(n, p)$?
3. What is the expected number of K^r -subgraphs in $G \in \mathcal{G}(n, p)$?
4. Characterize the graphs that occur as a subgraph in every graph of sufficiently large average degree.

5. In the usual terminology of measure spaces (and in particular, of probability spaces), the phrase ‘almost all’ is used to refer to a set of points whose complement has measure zero. Rather than considering a limit of probabilities in $\mathcal{G}(n, p)$ as $n \rightarrow \infty$, would it not be more natural to define a probability space on the set of *all* finite graphs (one copy of each) and to investigate properties of ‘almost all’ graphs in this space, in the sense above?
6. Show that if almost all $G \in \mathcal{G}(n, p)$ have a graph property $\mathcal{P}_1$ and almost all $G \in \mathcal{G}(n, p)$ have a graph property $\mathcal{P}_2$, then almost all $G \in \mathcal{G}(n, p)$ have both properties, i.e. have the property $\mathcal{P}_1 \cap \mathcal{P}_2$.
7. Show that, for constant $p \in (0, 1)$, almost every graph in $\mathcal{G}(n, p)$ has diameter 2.
8. Show that, for constant $p \in (0, 1)$, almost no graph in $\mathcal{G}(n, p)$ has a separating complete subgraph.
9. Derive Proposition 11.3.1 from Lemma 11.3.2.
10. Show that for every graph H there exists a function $p = p(n)$ such that $\lim_{n \rightarrow \infty} p(n) = 0$ but almost every $G \in \mathcal{G}(n, p)$ contains an induced copy of H .
- 11.⁺ (i) Show that, for every $0 < \epsilon \leq 1$ and $p = (1 - \epsilon)(\ln n)n^{-1}$, almost every $G \in \mathcal{G}(n, p)$ has an isolated vertex.
 (ii) Find a probability $p = p(n)$ such that almost every $G \in \mathcal{G}(n, p)$ is disconnected but the expected number of spanning trees of G tends to infinity as $n \rightarrow \infty$.
 (Hint for (ii): A theorem of Cayley states that K^n has exactly n^{n-2} spanning trees.)
- 12.⁺ Given $r \in \mathbb{N}$, find a $c > 0$ such that, for $p = cn^{-1}$, almost every $G \in \mathcal{G}(n, p)$ has a K^r minor. Can c be chosen independently of r ?
13. Find an increasing graph property without a threshold function, and a property that is not increasing but has a threshold function.
- 14.⁻ Let H be a graph of order k , and let h denote the number of graphs isomorphic to H on some fixed set of k elements. Show that $h \leq k!$. For which graphs H does equality hold?
- 15.⁻ For every $k \geq 1$, find a threshold function for $\{G \mid \Delta(G) \geq k\}$.
- 16.⁻ For every $d \in \mathbb{N}$, determine the threshold function for the property of containing the d -dimensional cube (see Exercise 2, Ch. 1), and for the property of containing the complete graph K^d .
17. Does the property of containing *any* tree of order k (for $k \geq 2$ fixed) have a threshold function? If so, which? If not, why not?
18. Show that $t(n) = n^{-1}$ is also a threshold function for the property of containing any cycle.

19. Consider the terms A_0 and A_1 in the proof of Lemma 11.4.4, which are both functions of n . Recall that $\mathbb{P}[H' \cup H'' \subseteq G] = p^{2\ell}$ both for $H' \cap H'' = \emptyset$ and for $|H' \cap H''| = 1$.
 - (i) Show that $A_0 \not\rightarrow 0$ (while $A_1 \rightarrow 0$) as $n \rightarrow \infty$.
 - (ii) Explain the difference without doing any formal calculations.
- 20.⁺ Given a graph H , let $\mathcal{P}$ be the property of containing an induced copy of H . Show that $\mathcal{P}$ has no threshold function unless H is complete.

Notes

There are a number of monographs and texts on the subject of random graphs. The first comprehensive monograph was B. Bollobás, *Random Graphs*, Academic Press 1985. Another advanced monograph is S. Janson, T. Luczak & A. Ruciński, *Random Graphs*, Wiley 2000; this concentrates on areas developed since *Random Graphs* was published. E.M. Palmer, *Graphical Evolution*, Wiley 1985, covers material similar to parts of *Random Graphs* but is written in a more elementary way. Compact introductions going beyond what is covered in this chapter are given by B. Bollobás, *Modern Graph Theory*, Springer GTM 184, 1998, and by M. Karoński, *Handbook of Combinatorics* (R.L. Graham, M. Grötschel & L. Lovász, eds.), North-Holland 1995.

A stimulating advanced introduction to the use of random techniques in discrete mathematics more generally is given by N. Alon & J.H. Spencer, *The Probabilistic Method*, Wiley 1992. One of the attractions of this book lies in the way it shows probabilistic methods to be relevant in proofs of entirely deterministic theorems, where nobody would suspect it. Other examples for this phenomenon are Alon's proof of Theorem 5.4.1, or the proof of Theorem 1.3.4; see the notes for Chapters 5 and 1, respectively.

The probabilistic method had its first origins in the 1940s, one of its earliest results being Erdős's probabilistic lower bound for Ramsey numbers (Theorem 11.1.3). Lemma 11.3.2 about the properties $\mathcal{P}_{i,j}$ is taken from Bollobás's Springer text cited above. A very readable rendering of the proof that, for constant p , every first order sentence about graphs is either almost surely true or almost surely false, is given by P. Winkler, Random structures and zero-one laws, in (N.W. Sauer et al., eds.) *Finite and Infinite Combinatorics in Sets and Logic* (NATO ASI Series C **411**), Kluwer 1993.

Theorem 11.3.5 is due to P. Erdős and A. Rényi, Asymmetric graphs, *Acta Math. Acad. Sci. Hungar.* **14** (1963), 295–315. For further references about the infinite random graph R see the notes in Chapter 8.

The seminal paper on graph evolution is P. Erdős & A. Rényi, On the evolution of random graphs, *Publ. Math. Inst. Hungar. Acad. Sci.* **5** (1960), 17–61. This paper also includes Theorem 11.4.5 for balanced graphs. The generalization to unbalanced subgraphs was first proved by Bollobás in 1981; see Karoński's *Handbook* chapter. The fact that all 'non-trivial' increasing graph properties have a threshold function was proved by B. Bollobás and A.G. Thomason, Threshold functions, *Combinatorica* **7** (1987), 35–38.

There is another way of defining a random graph G , which is just as natural and common as the model we considered. Rather than choosing the